INCH-POUND

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MILITARY SPECIFICATION

CRANE, FLOOR, PORTABLE, TRESTLE TYPE 8,000 POUND CAPACITY

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers a trestle type portable floor crane with an 8,000 pound capacity. The crane shall be equipped with a manually operated chain hoist which slides long a trolley.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packing, Methods of
MIL-P-514	Plates, Identification, Instructions and Marking
MIL-T-704	Treatment and Painting of Material
MIL-C-7474	Casters, Industrial
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-C-83286	Coating Urethane, Aliphatic Isocyanate, for Aerospace Applications

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRS, Robins AFB, GA 31098-5609 using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3950

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STANDARDS

FEDERAL

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-601	Boxes, Wood, Cleated, Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations For Use on Drawings on Specifications, Standards, and Technical Documents
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-808	Finishes, materials, and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting Identification Marking, and Data Plate Standards
MIL-STD-1595	Qualification of Aircraft, Missile, and Aerospace Fusion Welders
MS 24380	Casters Industrial
MS 51336	Lunette Coupler, Drawbar, Ring

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Non-Government Publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Welding Qualifications

(Application for copies should be addressed to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York NY 10017).

AMERICAN WELDING SOCIETY (AWS)

Welding Qualifications AWS D1.1

(Application for copies should be addressed to American Welding Society, 2501 NW Seventh Street, Miami FL 33125).

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J534 Lubrication Fittings

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein(except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities, and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Materials. Materials used in the construction of the cranes shall be as specified herein. Materials not definitely specified herein shall be sound, of uniform quality and condition, and free of seams, cracks, and other defects which may adversely affect the appearance, strength, endurance, or wear resistance of the finished product. Materials shall not be hammered, filed, or otherwise treated to conceal defects therein. Materials shall conform in composition, heat treatment, and suitability to the standard practices of reputable manufacturers producing equipment of this type.

Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack, or scale with age or extremes of climatic conditions shall not be used.

3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3 Design. The crane shall be designed as a self-contained portable unit having a rated capacity of 8,000 pounds. The end use shall be to hoist aircraft engines of the F-104, F-106, C-130, and other model aircraft. The crane shall consists of a movable overhead carriage for supporting a manually operated chain hoist, supporting members on each side to form a general trestle type frame, casters, floor brakes, and towbars for towing crane. When disassembled for shipping, the height of the component parts shall not exceed 45 inches. The basic structural members of the crane shall be seamless or welded type steel tubing affording good welding qualities. Figure 1 is furnished to show configuration only.

3.1.3.1 Reliability. The crane, including all components, shall have a Mean-Time-Between-Failure (MTBF) of not less than 100 hours, demonstrated in accordance with Test Plan XVIIC of MIL-STD-781. A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. Definitions of reliability terms not included herein shall be as found in MIL-STD-781. In the event of a conflict between these documents, MIL-STD-781 shall take precedence. The crane shall be designed for a life expectancy of at least 20 years.

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3.1.3.2 Maintainability. The crane shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.
b. A minimum amount of time for assembly, disassembling, locating trouble sources, and maintaining the crane. Where practical, components shall be located for rapid inspection and recognition of potential failure.
c. Ease of adjusting, servicing and replacing parts and components.
d. Use of readily available standard tools and equipment for maintenance.
e. Maintenance with a minimum number of tools.
f. Mean corrective maintenance downtime of not greater than 1/2 hour.
g. Maximum corrective maintenance downtime of no greater than 1 1/2 hours.
h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 10 minutes.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.3.1) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All such parts, when subject to removal or adjustment, shall not be swagged, peened, staked, or otherwise permanently deformed.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3.7 Component protection. All space in which work is performed during operation, service, and maintenance of the crane shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel.

3.1.3.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the crane.

3.1.4 Environmental conditions. The crane shall be capable of satisfactory storage and operation under the following environmental conditions:

a. Full time exposure to temperatures ranging from -40°F to +125°F.
b. Full time exposure to relative humidities up to 100 percent.
c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 to 4.5 mm falling at a rate of 13 cm per hour.
d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0 ml/80cm^2/hour.
e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750 ft/min, concentration 0.5 g/ft^3.
f. Part-time (five percent of the time) exposure to blowing sand; velocity 5,700 ft/min, concentration 0.0773 g/ft^3.

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3.2 Performance. The crane shall have the specified working capacity (see 1.1) and shall be capable of withstanding a total suspended load (static) or 100 percent overload without permanent deformation of any part of the system.

The crane shall show no evidence of instability or permanent deformation under any conditions of loading set forth above.

3.2.1 Rated capacity. The crane shall be capable of straddling and lifting from the ground a test load of dimensions 5'8"W X 3'H X 12'L weighing 8000 pounds (minimum). There shall be no interference of the lower structure while positioned above the test load. The crane shall be rated for lifting a 8,000 pound load on a 3 percent slope (maximum). The crane must track behind the prime mover in a fully loaded configuration (8,000 pounds) and the crane design will impede:

a. Tendency to yaw, sway, or skid under operating and braking conditions and minimum stopping distances.
b. Tendency to tip, tilt or jackknife.

3.2.2 Base and frame. The base and frame shall be of fabricated steel. The assembled unit shall be capable of sustaining a 100 percent (static) overload without permanent deformation.

DIMENSION IDENTIFICATION	DIMENSION
Weight (net)	900 lbs (maximum)
Overall height	14 ft 4 in (±2 in)
Overall width	6 ft 11 in (±2 in)
Span between supporting members	5 ft 11 in (±2 in)
Lift height (floor to center of lift hook)	11 ft 11 in (±2 in)

3.2.3 Casters and wheels. The casters shall provide ease of maneuverability and tracking when the crane is operating under a full load. The casters shall be supplied with wheel and swivel locks. Casters shall conform to the requirements specified in MIL-C-7474 and MS 24380. Two casters shall be of the swivel type, Part Number MS 24380-12SU, and two shall be of the nonswivel type, Part Number MS 24380-12RU.

3.2.4 Towbar. The frame assembly shall be equipped with an adjustable (without tools) towbar designed for towing by vehicles having a pintle hook height of 10 inches to 30 inches. The towbar shall incorporate a lunette eye conforming to MS 51336. The towbar shall not be less than five or more than six feet in length and shall be easily removable or stowable in an 'out-of-the-way' position when not in use.

3.2.5 Hoisting mechanism. The hoisting mechanism for the crane shall have an endless chain drive. Antifriction bearings shall be used throughout the hoist. The mechanism shall have an overall mechanical advantage of not less than 200. The hoisting mechanism shall be equipped with safety devices to prevent slipping of the load and to require positive effort to lower the load.

3.2.6 Floor brake. Braking of the crane will be accomplished by a minimum of four leveling jacks. The braking capacity of the crane shall permit stable operation in all possible loading conditions. Brakes shall lock when engaged and require positive action for disengagement.

3.2.7 Trolley. The crane shall be equipped with a hand chain trolley capable of being operated from the side of the trestle. It shall not interfere with the load lifting operation. The trolley shall permit movement of the hoist 1 foot 6 inches minimum in either direction of the centerline while suspending a capacity load.

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3.2.8 Weight/weight distribution. The crane shall weigh less than 900 pounds. The weight distribution of the unit shall not be unbalanced or improperly distributed.

3.3. Finish and protective coatings.

3.3.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808 and/or MIL-T-704, except that commercial (see 6.3.1) components may be prepared in accordance with the manufacturer standard practice, provided it's compatible with the exterior finish paint.

3.3.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defect and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

NOTE

"Equal" is defined as having performance identical to or Superior to DuPont Imron when subjected to the following tests:

a. Tests in accordance with MIL-C-46168:

(1) Paragraph 4.3.17 Flexibility.
(2) Paragraph 4.3.23 Accelerated weathering.

b. Tests in accordance with MIL-C-83286:

(1) Paragraph 4.8.6 Low temperature resistance.
(2) Paragraph 4.8.10 Resistance to taping.
(3) Paragraph 4.8.12 Pencil hardness.

3.3.2.2 Desert Sand. When specified (see 6.2), the surface of the crane shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595.

3.3.3 Rustproofing. Unless otherwise specified (see 6.2), the crane shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.4 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc. shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.4.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the crane in a readily accessible location. The identification shall contain the following information:

Nomenclature
Manufacturer's model number
Manufacturer's serial number
Manufacturer's name
Manufacturer's address
Manufacturer's service phone number
Date of manufacturer
Contract number
National stock number

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3.4.2 Control marking. All controls used in the operation of the crane shall be identified by nameplates securely installed in clearly visible locations.

3.4.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise, and adequate to enable operation of the crane without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.4.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.4.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Shipping weight.
b. Loading cubage.
c. Overall length, width, and height.

3.5 Part number of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known commercial parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair).

3.6 Welders and welding.

3.6.1 Welders. All welders employed in the fabrication of the crane, except commercial (see 6.3.1) components, shall be certified in accordance with the applicable requirements of MIL-STD-1595, the American Welding Society, AWS D1.1.

3.6.1.1 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

3.6.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.6.3 Screw assemblies. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.6.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.6.5 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

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3.6.6 Cleaning. The crane shall be cleaned. Loose, spattered, or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound, and other foreign matter shall be removed during and after final assembly.

3.7 First Article. The contractor shall furnish one crane for first article tests (See 4.4).

3.8 Workmanship. The crane, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners; etc.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of the inspection requirements as specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection and testing of the crane shall be classified as follows:

a. First article testing (see 4.4).
b. Acceptance testing (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.

4.3.3 Preparation for tests. The crane, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

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4.3.4 Test rejection criteria. Throughout all tests specified herein, the crane shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any lubricant, under any operating condition, except as allowed herein.
c. Structural failure of any component, including permanent deformation, or evidence of impending failure.
d. Evidence of excessive wear.
e. Interference between the crane components or between the crane, the ground, and all required obstacles, with the exception of normal contact by the casters.
f. Misalignment of components.
g. Conditions which present a safety hazard to personnel during operation, service, or maintenance.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples may be preproduction items or production items which meet the requirements of this specification. The samples shall be identical to the items procured under the terms of the contract. Approval of the first article test samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. One first article sample shall be produced and tested in accordance with contractor test sample test requirements (see 4.4.2.1).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.6.1 through 4.6.10.

4.4.3 Notification of tests. The procuring activity shall be notified of the time and location of all first article tests so that representatives of the procuring activity may attend. The notification shall be in writing and shall include the nomenclature and the national stock number.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).
b. Random sample testing (see 4.5.2).

All defects identified shall be corrected prior to acceptance.

4.5.1 Individual testing. Each production crane/hoist shall be subjected to the tests specified in 4.6.1, 4.6.2, and 4.6.3.

4.5.2 Rejection and retest. When a crane fails to meet the requirements of this specification, hoists to be delivered pursuant to the contract may not be accepted until the cause of the failure is determined, corrective action is taken by the contractor, and the component that failed is successfully retested to assure compliance with this specification. Tests not involving the failed component may be continued pending the investigation of a test failure.

4.6 Test methods.

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4.6.1 Examination of product. The crane shall be inspected to determine compliance with requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and their application, welding and riveting and markings. Any painted surface on the exterior of the crane may be tested in accordance with paragraph 4.8.10 of MIL-C-83286. For the first article and random sample cranes, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Operational checkout. All controls and levers shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation. These actions are to result in the operation of the following systems, but will not be limited to:

a. Operation and accessories.
b. Lift mechanisms, all attitudes.
c. Aircraft interface/pintle interface/tractor interface.
d. Trolley movement.
e. Miscellaneous equipment.

4.6.3 Capacity test. The crane shall be load tested to verify conformance with paragraph 3.2.1. The crane shall show no evidence of permanent deformation of any part of the system. There shall not be evidence of instability or mobility when leveling jacks are in place under capacity conditions.

4.6.4 Proof Test. The crane shall be capable of suspending a load of 200 percent capacity for not less than 30 minutes, with the following items checked:

a. Maximum 1/2 inch slippage downward.
b. No permanent deformation of any part of the system, after downloading.

4.6.5 Trolley Test. The trolley shall be tested to verify conformance with paragraph 3.2.7.

4.6.6 Hoisting Test. The crane shall be tested to verify conformance with paragraph 3.2.5.

4.6.7 Mobility Test. Under the conditions of paragraph 3.2.1 the crane shall track behind the prime mover for not less than 300 feet of continuous travel.

4.6.8 Brake tests. The leveling jacks shall be tested to ensure conformation with paragraph 3.2.6.

4.6.9 Design verification.

4.6.9.1 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and fully loaded conditions, to demonstrate compliance with paragraph 3.2.8 and 3.2.3.

4.6.9.2 Dimensional Verification. The dimensions specified herein shall be measured with the crane standing on a flat level surface. Overall height and crane lean shall be measured in both the empty and the fully loaded conditions, to demonstrate compliance with 3.2.2.

4.6.10 Environmental tests.

4.6.10.1 Low temperature storage and operation. The complete crane, including all accessories, shall be tested in accordance with Test method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.6.6.5. "Standard ambient" and controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the crane shall be exposed to the minimum required cold storage temperature of 3.1.4(a) in a steady state condition for 24 hours. For Procedure II, the crane shall be exposed to the minimum required cold operational temperature of 3.1.4 (a) in a steady state condition.

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4.6.10.2 High temperature storage and operation. The complete crane, including all accessories, shall be tested in accordance with Test Method 501.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with 4.6.2. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the crane shall be exposed to the maximum required hot storage temperature of 3.1.4(a) in a steady state condition for 24 hours. For Procedure II, the crane shall be exposed to the maximum required hot operational temperature of 3.1.4(a) in a steady state condition.

4.6.10.3 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with 3.1.4:

a. Humidity - Method 507.2, Procedure I.
b. Salt Fog - Method 509.2, Procedure I.
c. Sand and Dust - Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1,750 ft/min and high temperature adjustment shall be 125°F. For Procedure II, air velocity shall be 5,700 ft/min; sand concentration of 0.0623 g/ft^3, plus or minus 0.015 g/ft^3; 90 minutes per face duration.

4.6.11 Maintainability demonstration. Compliance with the maintenance downtime requirements shall be demonstrated by task performance. All scheduled maintenance tasks shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation, or miles traveled. The capability of performing the maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.7 Refurbishment. After successful complete of all the required testing, each crane shall be restored to like new condition.

5. PACKAGING.

5.1 Preservation. The preservation shall be Level A or commercial, as specified (see 6.2).

5.1.1 Level A. Unpainted ferrous metal surfaces, including threaded surfaces, shall be coated with Type P-1 preservative conforming to the applicable specifications listed in, and applied in accordance with MIL-P-116. The chain hoist, trolley, and all bearings of the casters shall be lubricated as for service. Casters, bolts, nut, washers, and any other small component parts requiring additional protection, shall be in accordance with MIL-P-116 for Method IC protection.

5.1.2 Commercial. The complete gantry shall be preserved in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be Level A, Level B, or commercial, as specified (see 6.2).

5.2.1 Level A. The larger structural components for each gantry shall be nested, arranged, and secured with steel strapping and bolting in combination with suitable wood block or battens as required to form compact nonshifting bundles. Each bundle shall be strapped girthwise with 0.035 by 1 1/4 inch flat steel strapping conforming to QQ-S-781, Class 1, Type I or IV, Finish B. A minimum of three straps shall be provided for each bundle. Strapping shall be spaced not to exceed approximately 36 inches on center, with end strapping placed not more than 18 inches from each end. The strapping shall be stapled to any wood blocking or battens provided. Metal edge protectors shall be used where strapping bears on sharp metal edges. The chain hoist, trolley, preserved and packaged items, and any other small unpackaged component

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parts not suitable for bundling shall be packed in close fitting boxes conforming to PPP-B-621, Class 2, or PPP-B-601, overseas type. The contents shall be blocked, braced to prevent movement within the boxes.

5.2.2 The complete gantry shall be packed as specified for Level A, except that the boxes shall conform to PPP-B-621, Class 1, or PPP-B-601, domestic type.

5.2.3 Commercial. The complete gantry shall be packed in a manner which will ensure arrival at destination in satisfactory condition. Packing shall comply with applicable carrier rules and regulations.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The crane covered by this specification is intended for hoisting aircraft engines of the F-104, F-106, C-130 and other model aircraft.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.3.2.1).
c. Polyurethane paint required, Desert Sand, Color Number 30313 of FED-STD-595 (see 3.3.2.2).
d. Rustproofing shall be performed in accordance with MIL-STD-1223 (see 3.3.3).
e. Preservation level, if not level A (see 4.8, 5.1).
f. Packing level, if not level A (see 5.2).

6.3 Definitions.

6.3.1 Commercial. A manufacturer's standard (product) having been marketed, in production, and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes, or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items for foreign military sales.

6.4 Data requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
	DI-A-7088	Conference Agenda	See DD Form 1423
	DI-A-7089	Conference Minutes	See DD Form 1423
	DI-A-4901	Test Procedures	See DD Form 1423
6.5	DI-A-4902	Test Report	See DD Form 1423
4.4.3	DI-T-23731A/T	Notification of Tests	See DD Form 1423
4.6.13	UDI-A-23122A/T	Shock Dynamic Design Analysis Report	See DD Form 1423
	DI-A-3011	Still Photo Coverage	See DD Form 1423

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-1, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

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6.5 Test report. Upon completion of the first article tests, a first article test report shall be prepared in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets and the report. The following shall be incorporated as a part of the test report:

a. All failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedure, as appropriate. Test conditions at the time of failures or irregular operation identified shall be recorded.
b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.
c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.
d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.
e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.
f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.
g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.
h. Test report for each component covered by a document listed in the DoDISS.
i. Plan for testing production units.

6.6 Subject term (key word) listing.

Casters
Collapsible
Chain
Hoist
Towbar
Trolley

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3950-F013

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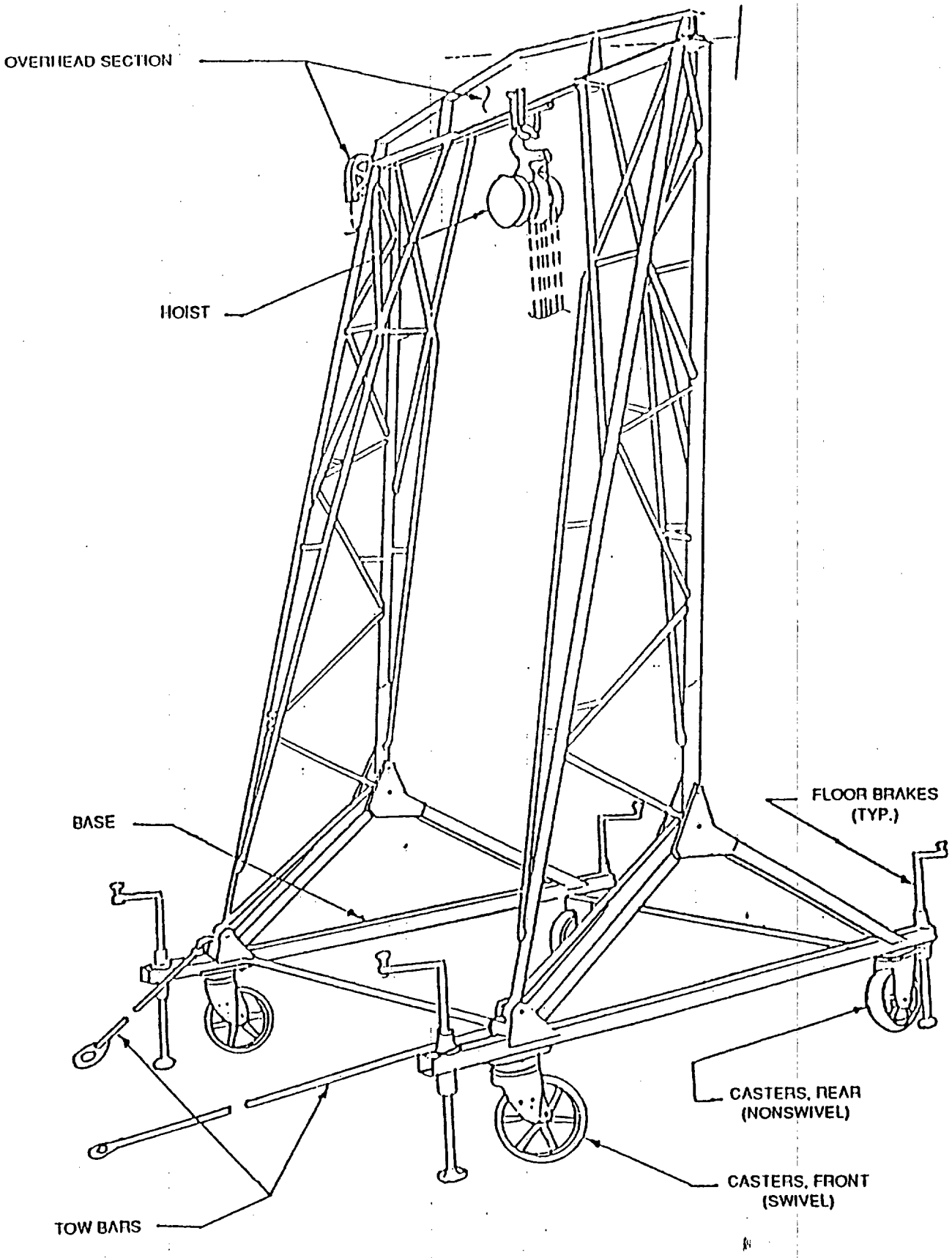


Figure 1. Crane, Floor, Portable, Trestle Type, 8,000 Pound Capacity (Typical)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-83811 (84)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: 912-926-5616 (2) AUTOVON: 468-5616	
c. ADDRESS (Include Zip Code) WR-ALC/MMVRS Robins AFB GA 31098-5365	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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